

Server Requirements

ASP.NET Core 8 Web API on IIS | IT Infrastructure Document

Operating System

Requirement	Specification
OS	Windows Server 2019 or 2022 (Recommended)
Architecture	64-bit only

Hardware Requirements

Component	Minimum	Recommended
CPU	2 Cores	4 Cores
RAM	4 GB	8 GB
Disk	40 GB HDD	80 GB SSD
Network	100 Mbps	1 Gbps

Software to Install

Software	Version	Notes
IIS (Web Server Role)	Built-in	Enable via Server Manager
.NET 8 Hosting Bundle	8.0.x (Latest)	Required for ASP.NET Core
Oracle Client	19c or 21c	For Oracle DB connectivity

IIS Role Services Required

Role Service	Required
Web Server (IIS)	YES
ASP.NET 4.8	YES
.NET Extensibility 4.8	YES
ISAPI Extensions	YES
ISAPI Filters	YES
IIS Management Console	YES

Static Content	YES
Default Document	YES
HTTP Errors	YES

Network / Firewall Requirements

Direction	Host / IP	Port	Purpose
Inbound	Any	8082	API access from client apps
Outbound	10.1.1.204	1527	Oracle DMS Database
Outbound	10.1.1.23	1526	Oracle SCM Database
Outbound	10.1.1.22	1526	Oracle HRMS Database
Outbound	newapi.ipca.com	443	LDAP / AD Authentication API
Outbound	EmailRelaySrv.ipca.com	587	SMTP Email Service

Application Configuration

Setting	Value
Application Type	ASP.NET Core 8 Web API
Deployment Folder	D:\Apps\AdminApi (or as per IT standard)
Port	8082 (matches existing Spring Boot port)
IIS App Pool CLR Version	No Managed Code
Authentication	JWT Bearer Token (no Windows Auth needed)
HTTPS	Recommended — configure SSL certificate in IIS

Important Notes for Server Team

1. App Pool must be set to 'No Managed Code' — this is specific to .NET Core apps and differs from classic ASP.NET.
2. The server requires outbound access to newapi.ipca.com on port 443 for Active Directory (AD) user authentication.
3. All three Oracle DB servers must be reachable from this server on their respective ports.
4. After installing .NET 8 Hosting Bundle, run 'iisreset' from an elevated command prompt.
5. Windows Authentication is NOT required — the API uses JWT tokens for all authentication.
6. If SSL is configured, update appsettings.json CORS origins to use https://.